

# Phish in Sheep's Clothing: Exploring the Authentication Pitfalls of Browser Fingerprinting

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# Phish in Sheep's Clothing: Exploring the Authentication Pitfalls of Browser Fingerprinting

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Abstract:

As users navigate the web they face a multitude of threats; among them, attacks that result in account compromise can be particularly devastating. In a world fraught with data breaches and sophisticated phishing attacks, web services strive to fortify user accounts by adopting new mechanisms that identify and prevent suspicious login attempts. More recently, browser fingerprinting techniques have been incorporated into the authentication workflow of major services as part of their decision-making process for triggering additional security mechanisms (e.g., two-factor authentication).

In this paper we present the first comprehensive and in-depth exploration of the security implications of real-world systems relying on browser fingerprints for authentication. Guided by our investigation, we develop a tool for automatically constructing fingerprinting vectors that replicate the process of target websites, enabling the extraction of fingerprints from users' devices that exactly match those generated by target websites. Subsequently, we demonstrate how phishing attackers can replicate users' fingerprints on different devices to deceive the risk-based authentication systems of high-value web services (e.g., cryptocurrency trading) to completely

bypass two-factor authentication. To gain a better understanding of whether attackers can carry out such attacks, we study the evolution of browser fingerprinting practices in phishing websites over time. While attackers do not generally collect all the necessary fingerprinting attributes, unfortunately that is not the case for attackers targeting certain financial institutions where we observe an increasing number of phishing sites capable of pulling off our attacks. To address the significant threat posed by our attack, we have disclosed our findings to the vulnerable vendors.

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